

```
In [1]: def multiply_for(a, b):
        """Multiply two whole numbers by repeated addition, using a for loop."""
        total = 0
        for _ in range(b):
            total = total + a
        return total
```

```
In [2]: def multiply_while(a, b):
        """Multiply two whole numbers by repeated addition, using a while loop."""
        total = 0
        count = 0
        while count < b:
            total = total + a
            count = count + 1
        return total
```

The while loop terminates when `count < b`, each iteration checks for that before running, so assuming `b` is not negative `count` will reach `b` because it starts as 0 and increases by 1 each pass.

```
In [3]: a, b = 7, 6

result_for = multiply_for(a, b)
result_while = multiply_while(a, b)
real_product = a * b

print(f"multiply_for({a}, {b}) = {result_for}")
print(f"multiply_while({a}, {b}) = {result_while}")
print(f"real product ({a} * {b}) = {real_product}")

assert result_for == real_product, "for-loop version disagrees with real product"
assert result_while == real_product, "while-loop version disagrees with real product"
assert result_for == result_while, "for-loop and while-loop versions disagree"

multiply_for(7, 6) = 42
multiply_while(7, 6) = 42
real product (7 * 6) = 42
```

```
In [4]: a, b = 0, 2

result_for = multiply_for(a, b)
result_while = multiply_while(a, b)
real_product = a * b

print(f"multiply_for({a}, {b}) = {result_for}")
print(f"multiply_while({a}, {b}) = {result_while}")
print(f"real product ({a} * {b}) = {real_product}")

assert result_for == real_product, "for-loop version disagrees with real product"
assert result_while == real_product, "while-loop version disagrees with real product"
assert result_for == result_while, "for-loop and while-loop versions disagree"
```

```
multiply_for(0, 2)    = 0
multiply_while(0, 2) = 0
real product (0 * 2) = 0
```

```
In [5]: a, b = 3, 0

result_for = multiply_for(a, b)
result_while = multiply_while(a, b)
real_product = a * b

print(f"multiply_for({a}, {b})    = {result_for}")
print(f"multiply_while({a}, {b}) = {result_while}")
print(f"real product ({a} * {b}) = {real_product}")

assert result_for == real_product, "for-loop version disagrees with real product"
assert result_while == real_product, "while-loop version disagrees with real product"
assert result_for == result_while, "for-loop and while-loop versions disagree"

multiply_for(3, 0)    = 0
multiply_while(3, 0) = 0
real product (3 * 0) = 0
```